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**WA Exams Practice Paper A, 2015**

**Question/Answer booklet**

**MATHEMATICS  
METHODS  
UNIT 1**

**Section One:  
Calculator-free**

**SOLUTIONS**

Student number: In figures

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In words

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Your name

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**Time allowed for this section**

Reading time before commencing work: five minutes  
Working time for section: fifty minutes

**Materials required/recommended for this section**

***To be provided by the supervisor***

This Question/Answer booklet  
Formula sheet

***To be provided by the candidate***

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

**Important note to candidates**

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

**Structure of this paper**

| Section                         | Number of questions available | Number of questions to be answered | Working time (minutes) | Marks available | Percentage of exam |
|---------------------------------|-------------------------------|------------------------------------|------------------------|-----------------|--------------------|
| Section One: Calculator-free    | 8                             | 8                                  | 50                     | 52              | 35                 |
| Section Two: Calculator-assumed | 13                            | 13                                 | 100                    | 98              | 65                 |
| <b>Total</b>                    |                               |                                    |                        |                 | 100                |

**Instructions to candidates**

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.
4. Additional working space pages at the end of this Question/Answer booklet are for planning or continuing an answer. If you use these pages, indicate at the original answer, the page number it is planned/continued on and write the question number being planned/continued on the additional working space page.
5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you do not use pencil, except in diagrams.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

## Section One: Calculator-free

35% (52 Marks)

This section has **eight (8)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time for this section is 50 minutes.

## Question 1

(5 marks)

(a) Solve  $2x = 5x^2$ .

(2 marks)

$$5x^2 - 2x = 0$$

$$x(5x - 2) = 0$$

$$x = 0 \text{ or } x = \frac{2}{5}$$

(b) Solve  $(x - 3)(x + 2) = 4x$ .

(3 marks)

$$x^2 - x - 6 = 4x$$

$$x^2 - 5x - 6 = 0$$

$$(x + 1)(x - 6) = 0$$

$$x = -1 \text{ or } x = 6$$

## Question 2

(4 marks)

- (a) Evaluate  $\frac{12!}{5! \times 8!}$ .

(2 marks)

$$\frac{12 \times 11 \times 10 \times 9}{120} \times \frac{8!}{8!} = 99$$

- (b) Determine the sum of  $\binom{6}{0} + \binom{6}{1} + \binom{6}{2} + \binom{6}{3} + \binom{6}{4} + \binom{6}{5} + \binom{6}{6}$ .

(2 marks)

$$\begin{aligned} \Sigma &= 2^6 \\ &= 64 \end{aligned}$$

Question 3

(7 marks)

- (a) A line has equation  $5x + 2y = 8$ . What is the gradient of this line?

(1 mark)

$$2y = -5x + 8$$

$$m = -\frac{5}{2}$$

- (b) Is the line  $3x + y = 0.5$  parallel, perpendicular, or neither, to the line  $3x + y = -2$ ? Justify your answer.

(1 mark)

Parallel, as both lines have the same gradient of  $-\frac{1}{3}$

- (c) Two points have coordinates P(5, -8) and Q(11, -20).

- (i) Determine the coordinates of the mid-point of P and Q.

(1 mark)

$$(8, -14)$$

- (ii) If Q is the mid-point of P and R, determine the coordinates of R.

(2 marks)

$$R(11+6, -20-12)$$

$$R(17, -32)$$

- (iii) Find the equation of the straight line passing through P and Q.

(2 marks)

$$m = \frac{-8 - -20}{5 - 11} = \frac{12}{-6} = -2$$

$$y = -2x + c$$

$$-8 = -2(5) + c$$

$$c = 2$$

$$y = -2x + 2$$

## Question 4

(7 marks)

(a) A quadratic function is given by  $f(x) = (x+1)^2 - 4$ . For this function, determine

(i) the coordinates of the  $y$ -intercept.

(1 mark)

$$(0, -3)$$

(ii) the equation of the line of symmetry.

(1 mark)

$$x = -1$$

(iii) the coordinates of the turning point.

(1 mark)

$$(-1, -4)$$

(b) Another quadratic function is given by  $y = 2 + 1.75x - 0.25x^2$ . Determine

(i) the equation of the line of symmetry.

(1 mark)

$$x = \frac{-1.75}{2(-0.25)}$$
$$x = \frac{7}{2}$$

(ii) the coordinates of the  $x$ -intercepts.

(3 marks)

$$-4(2 + 1.75x - 0.25x^2) = 0$$

$$x^2 - 7x - 8 = 0$$

$$(x+1)(x-8) = 0$$

$$(-1, 0) \text{ and } (8, 0)$$

Question 5

(8 marks)

- (a) A function is defined as  $f(x) = 10 - 2x$  over the domain  $\{x : x = 1, 3, 5\}$ . Determine the range of  $f(x)$ . (1 mark)

$$\{y : y = 0, 4, 8\}$$

- (b) State the natural domain and corresponding range for the function  $g(x) = 4 - \sqrt{x+3}$ . (2 marks)

$$\{x : x \geq -3\}$$

$$\{y : y \leq 4\}$$

- (c) The function  $h$  is given by  $h(x) = 2(x+3)^2 - 1$ .

- (i) Determine the  $x$ -coefficient of the expanded form of this polynomial. (1 mark)

$$2(x^2 + 6x + 9) - 1$$

$$\text{Ans: } 12$$

- (ii) State the range of the graph of  $y = 3h(x)$ . (1 mark)

$$y \geq -3$$

- (d) Comment on the behaviour of the following graphs as  $x \rightarrow -\infty$ .

- (i)  $y = x^5$ . (1 mark)

$$y \rightarrow -\infty$$

- (ii)  $y = x^{-1}$ . (1 mark)

$$y \rightarrow 0$$

- (iii)  $y = \sqrt{2-x}$ . (1 mark)

$$y \rightarrow \infty$$

## Question 6

(8 marks)

(a) A polynomial is given by  $5 - x + 2x^2 - 4x^3 + x^4$ .

(i) State the degree of this polynomial.

(1 mark)

4

(ii) Determine the sum of all the coefficients of this polynomial.

(1 mark)

$5 - 1 + 2 - 4 + 1 = 3$

(b) A row of Pascal's triangle starts with the numbers 1, 5, 10, ...

(i) Write down the numbers that complete the row.

(1 mark)

10, 5, 1

(ii) Express the sum of all the numbers in the row as a power of 2.

(1 mark)

$2^5$

(iii) Expand  $(2x - 1)^5$ .

(4 marks)

$$\begin{aligned}(2x - 1)^5 &= 1(2x)^5 + 5(2x)^4(-1) + 10(2x)^3(-1)^2 + 10(2x)^2(-1)^3 + 5(2x)(-1)^4 + 1(-1)^5 \\ &= 32x^5 - 80x^4 + 80x^3 - 40x^2 + 10x - 1\end{aligned}$$



Question 7

(6 marks)

- (a) Determine the exact value of  $\tan\left(-\frac{\pi}{3}\right)$ .

(1 mark)

$$-\sqrt{3}$$

- (b) Solve  $\sin^2(x) - \sin(x) = 2$  for  $0 \leq x \leq 360^\circ$ .

(3 marks)

$$\begin{aligned}\sin^2(x) - \sin(x) - 2 &= 0 \\ (\sin x + 1)(\sin x - 2) &= 0 \\ \sin x &= -1 \\ x &= 270^\circ\end{aligned}$$

- (c) Using the identity  $\cos(x - y) = \cos(x) \cdot \cos(y) + \sin(x) \cdot \sin(y)$  and the substitutions  $x = 90 - A$  and  $y = B$  show that  $\sin(A + B) = \sin(A) \cdot \cos(B) + \cos(A) \cdot \sin(B)$ .

(2 marks)

$$\begin{aligned}\cos(x - y) &= \cos(x) \cdot \cos(y) + \sin(x) \cdot \sin(y) \\ \cos(90 - A - B) &= \cos(90 - A) \cdot \cos(B) + \sin(90 - A) \cdot \sin(B) \\ \cos(90 - (A + B)) &= \sin(A) \cdot \cos(B) + \cos(A) \cdot \sin(B) \\ \sin(A + B) &= \sin(A) \cdot \cos(B) + \cos(A) \cdot \sin(B)\end{aligned}$$

## Question 8

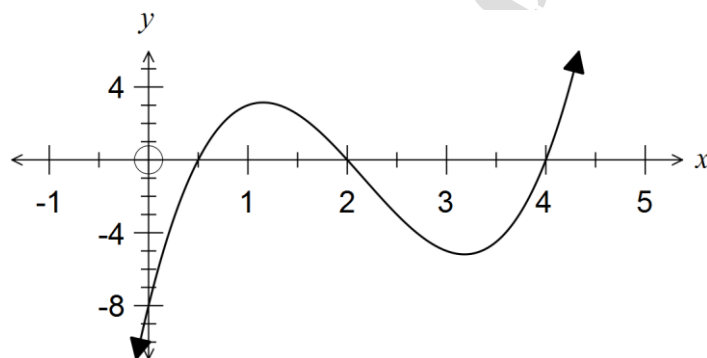
(7 marks)

- (a) Expand
- $(2x+1)(x+1)(2x-1)$
- .

(2 marks)

$$(4x^2 - 1)(x + 1) = 4x^3 + 4x^2 - x - 1$$

- (b) The graph of
- $y = 2x^3 - 13x^2 + 22x - 8$
- is shown below.

Factorise  $2x^3 - 13x^2 + 22x - 8$ .

(2 marks)

$$2(x-4)(x-2)(x-0.5) = (x-4)(x-2)(2x-1)$$

- (c) Solve
- $x^3 - 2x^2 - 5x + 6 = 0$
- .

(3 marks)

$$\begin{aligned} f(1) = 0 &\Rightarrow (x-1)(x^2 - x - 6) = 0 \\ (x-1)(x-3)(x+2) &= 0 \\ x = 1, x = 3, x = -2 \end{aligned}$$

Additional working space

Question number: \_\_\_\_\_

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